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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 26th February Evening Shift

JEE Main Question Paper with Answer Key

26 Feb 2021	90	15	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
Mathematics	22 questions
Physics	28 questions
Out of syllabus	15 questions
Answer key	at the end

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Match List - I with List - II.

Open solution

	List - I (Molecule)		List - II (Bond order)
(a)	Ne_2	(i)	1
(b)	N_2	(ii)	2
(c)	F_2	(iii)	0
(d)	O_2	(iv)	3

Choose the correct answer from the options given below :

(A) (a) $\rightarrow$ (i), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (iv)

(B) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

(C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)

(D) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)

Q2. MCQ (Single Correct Answer)

Ceric ammonium nitrate and $CHCl_3$/alc. KOH are used for the identification of functional groups present in _____ and _____ respectively.

Open solution

(A) amine, alcohol **(B)** alcohol, phenol **(C)** alcohol, amine **(D)** amine, phenol

Q3. MCQ (Single Correct Answer)

In $\overset{1}{C}H_2 = \overset{2}{C} = \overset{3}{C}H - \overset{4}{C}H_3$ molecule, the hybridization of carbon 1, 2, 3 and 4 respectively, are :

Open solution

(A) sp^2, sp, sp^2, sp^3

(B) sp^2, sp^2, sp^2, sp^3

(C) sp^3, sp, sp^3, sp^3

(D) sp^2, sp^3, sp^2, sp^3

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Q4. MCQ (Single Correct Answer)

A. Phenyl methanamine

B. N,N-Dimethylaniline

C. N-Methyl aniline

D. Benzenamine

Choose the correct order of basic nature of the above amines.

(A) D > C > B > A **(B)** A > C > B > D **(C)** D > B > C > A **(D)** A > B > C > D

Open solution

Q5. MCQ (Single Correct Answer)

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : In TlI_3, isomorphous to CsI_3, the metal is present in +1 oxidation state.

Reason R : Tl metal has fourteen f electrons in its electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is correct but R is not correct

(B) A is not correct but R is correct

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) Both A and R are correct and R is the correct explanation of A

Open solution

Q6. MCQ (Single Correct Answer)

Which of the following forms of hydrogen emits low energy β^- particles?

(A) Tritium ${}^{3}_{1}H$ **(B)** Proton H^+ **(C)** Protium ${}^{1}_{1}H$ **(D)** Deuterium ${}^{2}_{1}H$

Open solution

Q7. MCQ (Single Correct Answer)

Which pair of oxides is acidic in nature?

(A) CaO, SiO_2 **(B)** B_2O_3, CaO **(C)** B_2O_3, SiO_2 **(D)** N_2O, BaO

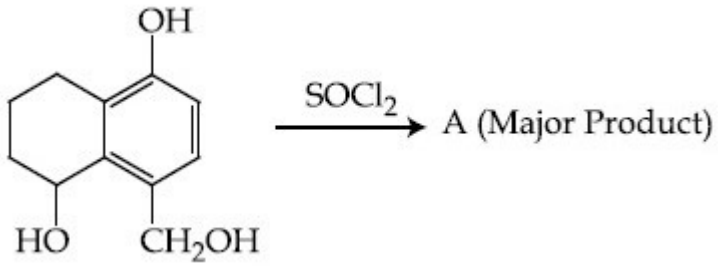
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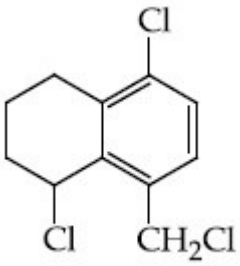
Q8. MCQ (Single Correct Answer)

Open solution

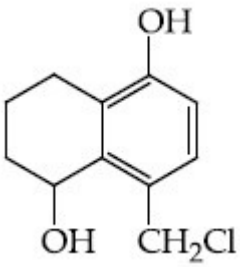
Identify A in the given reaction.



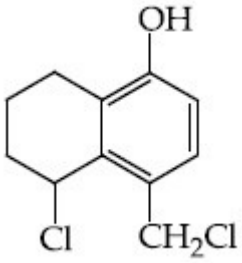
(A)



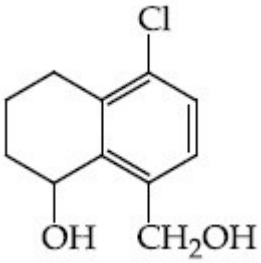
(B)



(C)



(D)

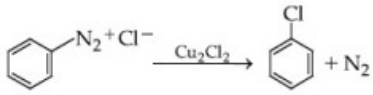
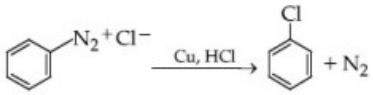


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Q9. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

List-I		List-II	
(a)		(i)	Wurtz reaction
(b)		(ii)	Sandmeyer reaction
(c)	$2CH_3CH_2Cl + 2Na \xrightarrow{Ether} C_2H_5 - C_2H_5 + 2NaCl$	(iii)	Fittig reaction
(d)	$2C_6H_5Cl + 2Na \xrightarrow{Ether} C_6H_5 - C_6H_5 + 2NaCl$	(iv)	Gatterman reaction

Choose the correct answer from the options given below :

(A) (a) → (ii), (b) → (iv), (c) → (i), (d) → (iii)

(B) (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)

(C) (a) → (iii), (b) → (i), (c) → (iv), (d) → (ii)

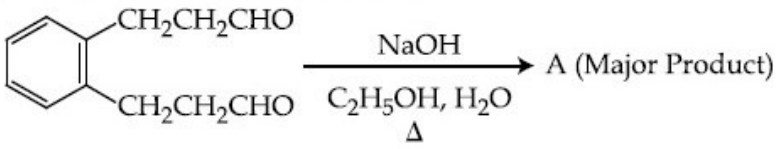
(D) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

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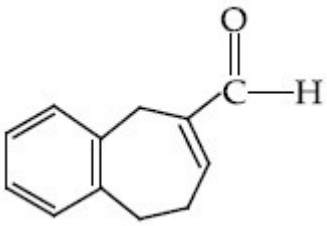
Q10. MCQ (Single Correct Answer)

Identify A in the following chemical reaction.

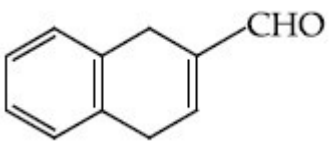
Open solution



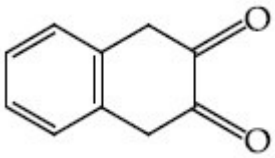
(A)



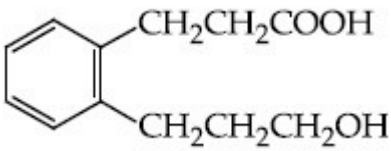
(B)



(C)



(D)



Q11. MCQ (Single Correct Answer)

2, 4-DNP test can be used to identify :

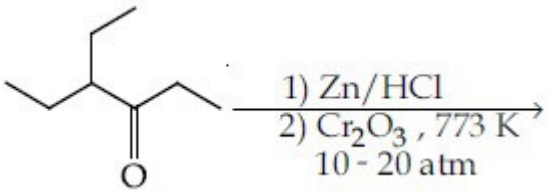
(A) ether (B) amine (C) aldehyde (D) halogens

Open solution

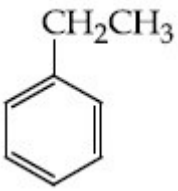
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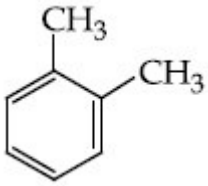
Q12. MCQ (Single Correct Answer)

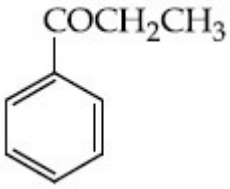
Open solution

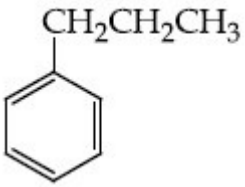


Considering the above reaction, the major product among the following is :

(A) 

(B) 

(C) 

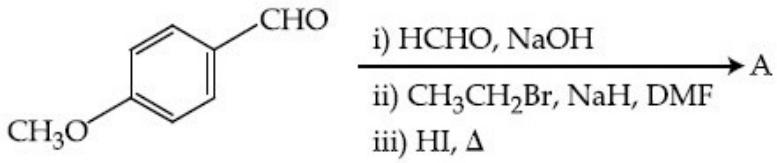
(D) 

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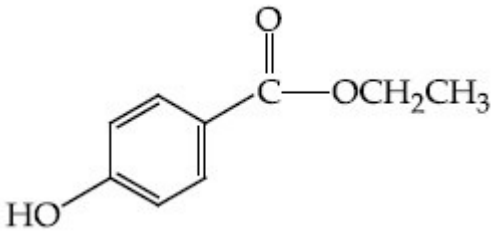
Q13. MCQ (Single Correct Answer)

Identify A in the given chemical reaction.

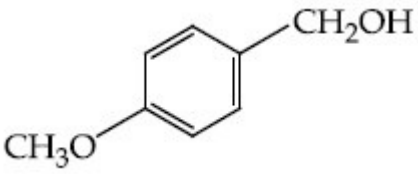
Open solution



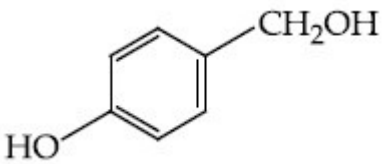
(A)



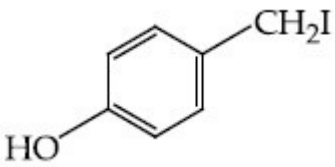
(B)



(C)



(D)



Q14. MCQ (Single Correct Answer)

The correct order of electron gain enthalpy is :

(A) Te > Se > S > O (B) S > Se > Te > O (C) S > O > Se > Te (D) O > S > Se > Te

Open solution

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Q15. MCQ (Single Correct Answer)

Match List - I with List - II.

	List - I		List - II
(a)	Sucrose	(i)	β-D-Galactose and β-D-Glucose
(b)	Lactose	(ii)	α-D-Glucose and β-D-Fructose
(c)	Maltose	(iii)	α-D-Glucose and α-D-Glucose

Choose the correct answer from the options given below :

(A) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (i)

(B) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii)

(C) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iii)

(D) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii)

Open solution

Q16. MCQ (Single Correct Answer)

Seliwanoff test and Xanthoproteic test are used for the identification of _____ and _____ respectively.

(A) ketoses, proteins

(B) proteins, ketoses

(C) aldoses, ketoses

(D) ketoses, aldoses

Open solution

Q17. Numerical

When 12.2 g of benzoic acid is dissolved in 100 g of water, the freezing point of solution was found to be -0.93°C ($K_f(H_2O)$ = 1.86 K kg mol^{-1}). The number (n) of benzoic acid molecules associated (assuming 100% association) is _____.

Open solution

Q18. Numerical

If the activation energy of a reaction is 80.9 kJ mol^{-1}, the fraction of molecules at 700 K, having enough energy to react to form products is e^{-x}. The value of x is _____. (Rounded off to the nearest integer) [Use R = 8.31 J K^{-1} mol^{-1}]

Open solution

Q19. Numerical

In mildly alkaline medium, thiosulphate ion is oxidized by MnO_4^- to "A". The oxidation state of sulphur in "A" is _____.

Open solution

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Q20. Numerical

The average S-F bond energy in kJ mol^{-1} of SF_6 is _____. (Rounded off to the nearest integer)

[Given : The values of standard enthalpy of formation of SF_6(g), S(g) and F(g) are - 1100, 275 and 80 kJ mol^{-1} respectively.]

Open solution

Q21. Numerical

Emf of the following cell at 298K in V is x $\times$ 10^{-2}.

$Zn|Zn^{2+}(0.1\ M)||Ag^{+}\ (0.01\ M)|Ag$

The value of x is _____. (Rounded off to the nearest integer)

[Given : $E^{\theta}_{Zn^{2+}/Zn} = -0.76V; E^{\theta}_{Ag^{+}/Ag} = +0.80V; \frac{2.303RT}{F} = 0.059$]

Open solution

Q22. Numerical

The pH of ammonium phosphate solution, if pk_a of phosphoric acid and pk_b of ammonium hydroxide are 5.23 and 4.75 respectively, is _____. (Nearest integer)

Open solution

Q23. Numerical

The number of stereoisomers possible for $[Co(ox)_2(Br)(NH_3)]^{2-}$ is _____. [ox = oxalate]

Open solution

Q24. Numerical

A ball weighing 10 g is moving with a velocity of 90 ms^{-1}. If the uncertainty in its velocity is 5%, then the uncertainty in its position is _____ $\times$ 10^{-33} m. (Rounded off to the nearest integer)

[Given : h = 6.63 $\times$ 10^{-34} Js]

Open solution

Q25. Numerical

The $NaNO_3$ weighed out to make 50 mL of an aqueous solution containing 70.0 mg Na^{+} per mL is _____ g. (Rounded off to the nearest integer)

[Given : Atomic weight in g mol^{-1} - Na : 23; N : 14; O : 16]

Open solution

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Mathematics 22 questions

Q26. MCQ (Single Correct Answer)

Let f(x) be a differentiable function at x = a with f'(a) = 2 and f(a) = 4.

Then $\lim_{x \to a} \frac{xf(a) - af(x)}{x - a}$ equals :

(A) 4 − 2a (B) 2a + 4 (C) a + 4 (D) 2a − 4

Open solution

Q27. MCQ (Single Correct Answer)

Let A(1, 4) and B(1, −5) be two points. Let P be a point on the circle $(x - 1)^2 + (y - 1)^2 = 1$ such that $(PA)^2 + (PB)^2$ have maximum value, then the points, P, A and B lie on :

(A) a straight line (B) an ellipse (C) a parabola (D) a hyperbola

Open solution

Q28. MCQ (Single Correct Answer)

Consider the following system of equations :

x + 2y − 3z = a

2x + 6y − 11z = b

x − 2y + 7z = c,

where a, b and c are real constants. Then the system of equations :

(A) has no solution for all a, b and c

(B) has a unique solution when 5a = 2b + c

(C) has infinite number of solutions when 5a = 2b + c

(D) has a unique solution for all a, b and c

Open solution

Q29. MCQ (Single Correct Answer)

A natural number has prime factorization given by $n = 2^x 3^y 5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}$, $y > z$. Then the number of odd divisors of n, including 1, is :

(A) 11 (B) 6 (C) 12 (D) 6x

Open solution

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Q30. MCQ (Single Correct Answer)

If 0 < a, b < 1, and $\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$, then the value of

$(a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \ldots\ldots$ is :

(A) $\log_e 2$ (B) e

(C) $\log_e\left(\frac{e}{2}\right)$ (D) $e^2 - 1$

Open solution

Q31. MCQ (Single Correct Answer)

Let $f(x) = \int_0^x e^t f(t)dt + e^x$ be a differentiable function for all x∈R. Then f(x) equals :

(A) $e^{(e^{x-1})}$ (B) $2e^{e^x} - 1$ (C) $2e^{e^x-1} - 1$ (D) $e^{e^x} - 1$

Open solution

Q32. MCQ (Single Correct Answer)

For x > 0, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)}dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to :

(A) $\frac{1}{2}$ (B) -1 (C) 0 (D) 1

Open solution

Q33. MCQ (Single Correct Answer)

Let $A = \{1, 2, 3, \ldots\ldots, 10\}$ and $f : A \to A$ be defined as

$$f(k) = \begin{cases} k+1 & if\ k\ is\ odd \\ k & if\ k\ is\ even \end{cases}$$

Then the number of possible functions $g : A \to A$ such that $gof = f$ is :

(A) 5^5 (B) 10^5 (C) 5! (D) ${}^{10}C_5$

Open solution

Q34. MCQ (Single Correct Answer)

Let A_1 be the area of the region bounded by the curves y = sinx, y = cosx and y-axis in the first quadrant. Also, let A_2 be the area of the region bounded by the curves y = sinx, y = cosx, x-axis and x = $\frac{\pi}{2}$ in the first quadrant. Then,

(A) $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$ (B) $A_1 = A_2$ and $A_1 + A_2 = \sqrt{2}$

(C) $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$ (D) $A_1 : A_2 = 1 : 2$ and $A_1 + A_2 = 1$

Open solution

Q35. MCQ (Single Correct Answer)

Let $f(x) = \sin^{-1}x$ and $g(x) = \frac{x^2-x-2}{2x^2-x-6}$. If $g(2) = \lim_{x\to 2} g(x)$, then the domain of the function fog is :

(A) $(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$ (B) $(-\infty, -2] \cup [-1, \infty)$

(C) $(-\infty, -2] \cup \left[-\frac{3}{2}, \infty\right)$ (D) $(-\infty, -1] \cup [2, \infty)$

Open solution

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Q36. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be defined as

$$f(x)=\begin{cases}2\sin\left(-\frac{\pi x}{2}\right), if\, x<-1\\ |ax^2+x+b|,\ if-1\le x\le 1\\ \sin(\pi x),\ if\, x>1\end{cases}$$ If f(x) is continuous on R, then a + b equals :

(A) -3 (B) 3 (C) -1 (D) 1

Open solution

Q37. MCQ (Single Correct Answer)

If the locus of the mid-point of the line segment from the point (3, 2) to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r, then r is equal to :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{3}$

Open solution

Q38. MCQ (Single Correct Answer)

If vectors $\overrightarrow{a_1}=x\hat{i}-\hat{j}+\hat{k}$ and $\overrightarrow{a_2}=\hat{i}+y\hat{j}+z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i}+y\hat{j}+z\hat{k}$ is :

(A) $\frac{1}{\sqrt{3}}\left(\hat{i}-\hat{j}+\hat{k}\right)$ (B) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\hat{i}-\hat{j}\right)$ (D) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}-\hat{k}\right)$

Open solution

Q39. MCQ (Single Correct Answer)

A seven digit number is formed using digits 3, 3, 4, 4, 4, 5, 5. The probability, that number so formed is divisible by 2, is :

(A) $\frac{1}{7}$ (B) $\frac{4}{7}$ (C) $\frac{6}{7}$ (D) $\frac{3}{7}$

Open solution

Q40. Numerical

The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is _____.

Open solution

Q41. Numerical

If the matrix $A=\begin{bmatrix}1&0&0\\0&2&0\\3&0&-1\end{bmatrix}$ satisfies the equation

$A^{20}+\alpha A^{19}+\beta A=\begin{bmatrix}1&0&0\\0&4&0\\0&0&1\end{bmatrix}$ for some real numbers α and β, then $\beta-\alpha$ is equal to _____.

Open solution

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Q42. Numerical

Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.

Open solution

Q43. Numerical

Let z be those complex numbers which satisfy

$|z + 5| \leq 4$ and $z(1 + i) + \overline{z}(1 - i) \geq -10$, $i = \sqrt{-1}$.

If the maximum value of $|z + 1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is _____.

Open solution

Q44. Numerical

If the arithmetic mean and geometric mean of the pth and q^{th} terms of the sequence $-16, 8, -4, 2, \ldots\ldots$ satisfy the equation $4x^2 - 9x + 5 = 0$, then p + q is equal to _____.

Open solution

Q45. Numerical

If $I_{m,n} = \int_0^1 x^{m-1}(1-x)^{n-1}dx$, for m, $n \geq 1$, and $\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx = \alpha I_{m,n} \alpha \in R$, then α equals _____.

Open solution

Q46. Numerical

Let $X_1, X_2, \ldots\ldots, X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is _____.

Open solution

Q47. Numerical

Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval (a, a + 1). Then, |a| is equal to _____.

Open solution

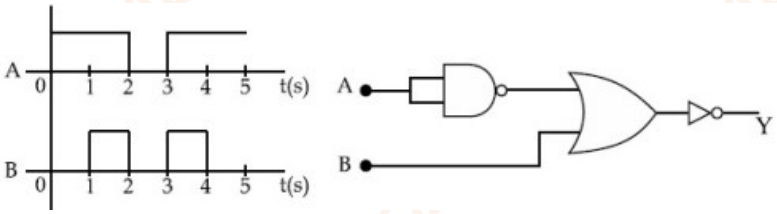
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Physics 28 questions

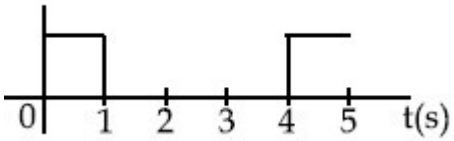
Q48. MCQ (Single Correct Answer)

Open solution

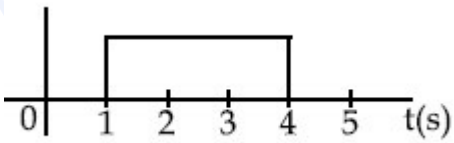
Draw the output signal Y in the given combination of gates.



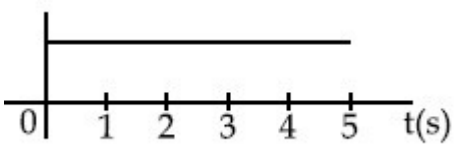
(A)



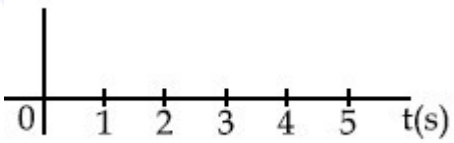
(B)



(C)



(D)



Q49. MCQ (Single Correct Answer)

Open solution

An aeroplane, with its wings spread 10 m, is flying at a speed of 180 km/h in a horizontal direction. The total intensity of earth's field at that part is 2.5×10^{-4} Wb/m^2 and the angle of dip is 60°. The emf induced between the tips of the plane wings will be _____.

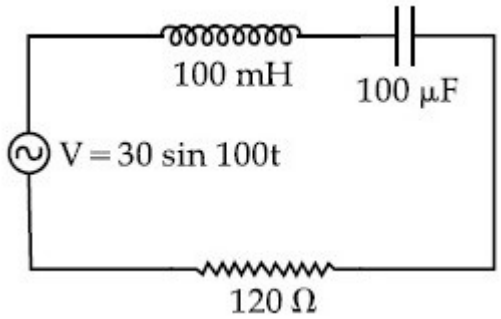
(A) 88.37 mV (B) 62.50 mV (C) 54.125 mV (D) 108.25 mV

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Q50. MCQ (Single Correct Answer)

Find the peak current and resonant frequency of the following circuit (as shown in figure).

Open solution



(A) 2 A and 100 Hz (B) 2 A and 50 Hz (C) 0.2 A and 100 Hz (D) 0.2 A and 50 Hz

Q51. MCQ (Single Correct Answer)

A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and comes to rest. The correct value of $\frac{t_1}{t_2}$ will be :

Open solution

(A) $\frac{a_1+a_2}{a_2}$ (B) $\frac{a_1+a_2}{a_1}$ (C) $\frac{a_2}{a_1}$ (D) $\frac{a_1}{a_2}$

Q52. MCQ (Single Correct Answer)

Given below are two statements:

Open solution

Statement I : An electric dipole is placed at the center of a hollow sphere. The flux of the electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement II : If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius r (< R) is zero but the electric flux passing through this closed spherical surface of radius r is not zero.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q53. MCQ (Single Correct Answer)

If 'C' and 'V' represent capacitance and voltage respectively then what are the dimensions of λ where C/V = λ ?

Open solution

(A) $[M^{-3}L^{-4}I^{3}T^{7}]$ (B) $[M^{-2}L^{-3}I^{2}T^{6}]$

(C) $[M^{-2}L^{-4}I^{3}T^{7}]$ (D) $[M^{-1}L^{-3}I^{-2}T^{-7}]$

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Q54. MCQ (Single Correct Answer)

The recoil speed of a hydrogen atom after it emits a photon in going from n = 5 state to n = 1 state will be :

(A) 4.34 m/s (B) 2.19 m/s (C) 3.25 m/s (D) 4.17 m/s

Open solution

Q55. MCQ (Single Correct Answer)

A wire of 1Ω has a length of 1 m. It is stretched till its length increases by 25%. The percentage change in resistance to the nearest integer is :

(A) 76% (B) 12.5% (C) 25% (D) 56%

Open solution

Q56. MCQ (Single Correct Answer)

The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. Then choose the correct relation for these vectors.

(A) $\vec{b} = \vec{a} + 2\vec{c}$

(B) $\vec{b} = \vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c}$

(C) $\vec{b} = 2\vec{a} + \vec{c}$

(D) $\vec{b} = \vec{a} - \vec{c}$

Open solution

Q57. MCQ (Single Correct Answer)

The internal energy (U), pressure (P) and volume (V) of an ideal gas are related as U = 3PV + 4. The gas is :

(A) either monoatomic or diatomic. (B) monoatomic only.

(C) polyatomic only. (D) diatomic only.

Open solution

Q58. MCQ (Single Correct Answer)

The length of metallic wire is l_1 when tension in it is T_1. It is l_2 when the tension is T_2. The original length of the wire will be :

(A) $\frac{T_1 l_1 - T_2 l_2}{T_2 - T_1}$ (B) $\frac{l_1 + l_2}{2}$

(C) $\frac{T_2 l_1 + T_1 l_2}{T_1 + T_2}$ (D) $\frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$

Open solution

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Q59. MCQ (Single Correct Answer)

Open solution

The trajectory of a projectile in a vertical plane is $y = \alpha x - \beta x^2$, where α and β are constants and x & y are respectively the horizontal and vertical distances of the projectile from the point of projection. The angle of projection θ and the maximum height attained H are respectively given by :

(A) $\tan^{-1}\alpha, \frac{\alpha^2}{4\beta}$

(B) $\tan^{-1}\alpha, \frac{4\alpha^2}{\beta}$

(C) $\tan^{-1}\left(\frac{\beta}{\alpha}\right), \frac{\alpha^2}{\beta}$

(D) $\tan^{-1}\beta, \frac{\alpha^2}{2\beta}$

Q60. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labeled as Assertion A and the other is labeled as Reason R.

Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and R are true and R is the correct explanation of A

(D) Both A and R are true but R is NOT the correct explanation of A

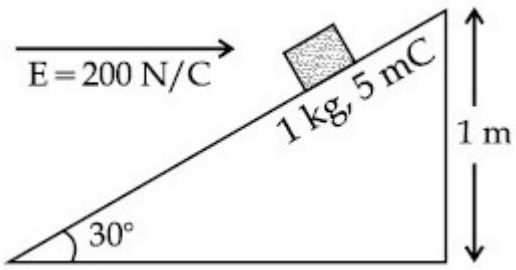
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Q61. MCQ (Single Correct Answer)

Open solution

An inclined plane making an angle of 30° with the horizontal is placed in a uniform horizontal electric field $200\frac{N}{C}$ as shown in the figure. A body of mass 1 kg and charge 5 mC is allowed to slide down from rest at a height of 1 m. If the coefficient of friction is 0.2, find the time taken by the body to reach the bottom.

[g = 9.8 m/s^2; $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$]

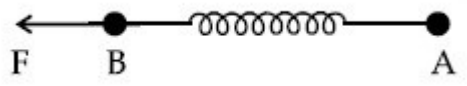


(A) 0.46 s (B) 0.92 s (C) 1.3 s (D) 2.3 s

Q62. MCQ (Single Correct Answer)

Open solution

Two masses A and B, each of mass M are fixed together by a massless spring. A force acts on the mass B as shown in the figure. If the mass A starts moving away from mass B with acceleration 'a', then the acceleration of mass B will be :



(A) $\frac{MF}{F+Ma}$ (B) $\frac{F+Ma}{M}$ (C) $\frac{Ma-F}{M}$ (D) $\frac{F-Ma}{M}$

Q63. MCQ (Single Correct Answer)

Open solution

A cord is wound round the circumference of wheel of radius r. The axis of the wheel is horizontal and the moment of inertia about it is I. A weight mg is attached to the cord at the end. The weight falls from rest. After falling through a distance 'h', the square of angular velocity of wheel will be :

(A) $\frac{2mgh}{I+2mr^2}$ (B) $\frac{2mgh}{I+mr^2}$ (C) 2gh (D) $\frac{2gh}{I+mr^2}$

Q64. MCQ (Single Correct Answer)

Open solution

A particle executes S.H.M., the graph of velocity as a function of displacement is :

(A) a parabola (B) a helix (C) an ellipse (D) a circle

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Q65. MCQ (Single Correct Answer)

A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340 Hz. When fork A is filed, the beat frequency decreases to 2 beats/s. What is the frequency of fork A?

(A) 335 Hz (B) 345 Hz (C) 338 Hz (D) 342 Hz

Open solution

Q66. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : A second's pendulum has a time period of 1 second.

Statement II : It takes precisely one second to move between the two extreme positions.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Open solution

Q67. Numerical

1 mole of rigid diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is _____. [R = universal gas constant]

Open solution

Q68. Numerical

A particle executes S.H.M. with amplitude 'a', and time period 'T'. The displacement of the particle when its speed is half of maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is _____.

Open solution

Q69. Numerical

Time period of a simple pendulum is T. The time taken to complete $\frac{5}{8}$ oscillations starting from mean position is $\frac{\alpha}{\beta}T$. The value of α is _____.

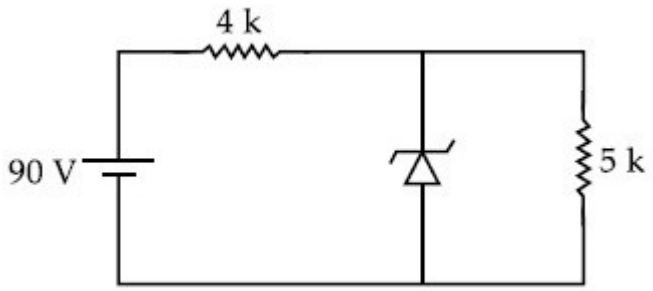
Open solution

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Q70. Numerical

The zener diode has a V_Z = 30V. The current passing through the diode for the following circuit is _____ mA.

Open solution



Q71. Numerical

The volume V of a given mass of monoatomic gas changes with temperature T according to the relation $V = KT^{\frac{2}{3}}$. The workdone when temperature changes by 90K will be xR. The value of x is _____. [R = universal gas constant]

Open solution

Q72. Numerical

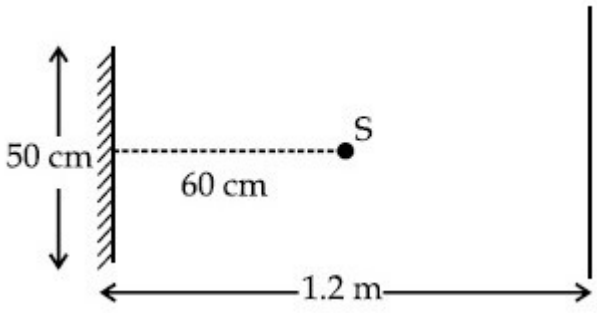
27 similar drops of mercury are maintained at 10V each. All these spherical drops combine into a single big drop. The potential energy of the bigger drop is _____ times that of a smaller drop.

Open solution

Q73. Numerical

A point source of light S, placed at a distance 60cm in front of the centre of a plane mirror of width 50 cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2 m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is _____ cm.

Open solution

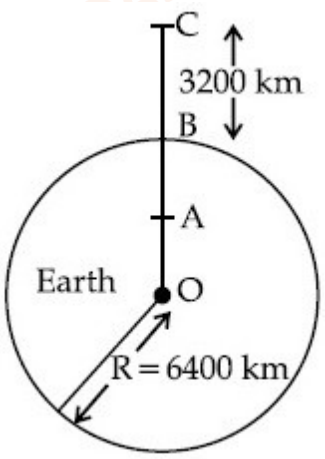


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Q74. Numerical

In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be x : y. The value of x is _____.

Open solution



Q75. Numerical

Two streams of photons, possessing energies equal to twice and ten times the work function of metal are incident on the metal surface successively. The value of ratio of maximum velocities of the photoelectrons emitted in the two respective cases is x : y. The value of x is _____.

Open solution

Out of syllabus 15 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II.

Open solution

	List - I		List - II
(a)	Sodium Carbonate	(i)	Deacon
(b)	Titanium	(ii)	Castner-Kellner
(c)	Chlorine	(iii)	van-Arkel
(d)	Sodium hydroxide	(iv)	Solvay

Choose the correct answer from the options given below :

(A) (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii)

(B) (a) → (iv), (b) → (i), (c) → (ii), (d) → (iii)

(C) (a) → (iii), (b) → (ii), (c) → (i), (d) → (iv)

(D) (a) → (i), (b) → (iii), (c) → (iv), (d) → (ii)

Q77. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II.

Open solution

	List - I		List - II
(a)	Siderite	(i)	Cu
(b)	Calamine	(ii)	Ca
(c)	Malachite	(iii)	Fe
(d)	Cryolite	(iv)	Al
		(v)	Zn

Choose the correct answer from the options given below :

(A) (a) → (i), (b) → (ii), (c) → (iii), (d) → (iv)

(B) (a) → (iii), (b) → (i), (c) → (v), (d) → (ii)

(C) (a) → (i), (b) → (ii), (c) → (v), (d) → (iii)

(D) (a) → (iii), (b) → (v), (c) → (i), (d) → (iv)

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Q78. MCQ (Single Correct Answer) · **Chemistry**

The nature of charge on resulting colloidal particles when $FeCl_3$ is added to excess of hot water is :

(A) neutral

(B) sometimes positive and sometimes negative

(C) positive

(D) negative

Open solution

Q79. MCQ (Single Correct Answer) · **Chemistry**

Calgon is used for water treatment. Which of the following statement is NOT true about Calgon?

(A) It is also known as Graham's salt.

(B) It is polymeric compound and is water soluble.

(C) It does not remove Ca^{2+} ion by precipitation.

(D) Calgon contains the 2nd most abundant element by weight in the Earth's crust.

Open solution

Q80. Numerical · **Chemistry**

The number of octahedral voids per lattice site in a lattice is _____. (Rounded off to the nearest integer)

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let $F_1(A, B, C) = (A \wedge \sim B) \vee [\sim C \wedge (A \vee B)] \vee \sim A$ and $F_2(A, B) = (A \vee B) \vee (B \rightarrow \sim A)$ be two logical expressions. Then :

(A) Both F_1 and F_2 are not tautologies

(B) F_1 and F_2 both are tautologies

(C) F_1 is not a tautology but F_2 is a tautology

(D) F_1 is a tautology but F_2 is not a tautology

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

If the mirror image of the point (1, 3, 5) with respect to the plane

$4x - 5y + 2z = 8$ is (α, β, γ), then $5(\alpha + \beta + \gamma)$ equals :

(A) 39 (B) 41 (C) 47 (D) 43

Open solution

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Q83. MCQ (Single Correct Answer) · **Mathematics**

Let slope of the tangent line to a curve at any point P(x, y) be given by $\frac{xy^2+y}{x}$. If the curve intersects the line x + 2y = 4 at x = -2, then the value of y, for which the point (3, y) lies on the curve, is :

(A) $-\frac{18}{19}$ (B) $-\frac{4}{3}$ (C) $\frac{18}{35}$ (D) $-\frac{18}{11}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

The triangle of maximum area that can be inscribed in a given circle of radius 'r' is :

(A) An equilateral triangle having each of its side of length $\sqrt{3}$r.

(B) An equilateral triangle of height $\frac{2r}{3}$.

(C) A right angle triangle having two of its sides of length 2r and r.

(D) An isosceles triangle with base equal to 2r.

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The sum of the series

$\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to :

(A) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$ (B) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(C) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$ (D) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let L be a line obtained from the intersection of two planes x + 2y + z = 6 and y + 2z = 4. If point P(α, β, γ) is the foot of perpendicular from (3, 2, 1) on L, then the value of 21($\alpha + \beta + \gamma$) equals :

(A) 102 (B) 142 (C) 136 (D) 68

Open solution

Q87. Numerical · **Mathematics**

Let the normals at all the points on a given curve pass through a fixed point (a, b). If the curve passes through (3, -3) and (4, $-2\sqrt{2}$), and given that a $-$ $2\sqrt{2}$ b = 3, then ($a^2 + b^2 + ab$) is equal to _____.

Open solution

Q88. Numerical · **Mathematics**

Let L be a common tangent line to the curves

$4x^2 + 9y^2 = 36$ and $(2x)^2 + (2y)^2 = 31$. Then the

square of the slope of the line L is _____.

Open solution

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Q89. MCQ (Single Correct Answer) · **Physics**

A radioactive sample is undergoing α decay. At any time t_1, its activity is A and another time t_2, the activity is $\frac{A}{5}$. What is the average life time for the sample?

(A) $\frac{\ln 5}{t_2 - t_1}$ (B) $\frac{\ln(t_2+t_1)}{2}$ (C) $\frac{t_1-t_2}{\ln 5}$ (D) $\frac{t_2-t_1}{\ln 5}$

Open solution

Q90. Numerical · **Physics**

If the highest frequency modulating a carrier is 5 kHz, then the number of AM broadcast stations accommodated in a 90 kHz bandwidth are _____.

Open solution

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Answer Key

Chemistry

1. **C**	2. **C**	3. **A**	4. **D**	5. **C**
6. **A**	7. **C**	8. **C**	9. **A**	10. **A**
11. **C**	12. **A**	13. **D**	14. **B**	15. **C**
16. **A**	17. **2**	18. **14**	19. **6**	20. **309**
21. **147**	22. **7**	23. **3**	24. **1**	25. **13**

Mathematics

26. **A**	27. **A**	28. **C**	29. **C**	30. **A**
31. **C**	32. **A**	33. **B**	34. **A**	35. **A**
36. **C**	37. **B**	38. **A**	39. **D**	40. **1000**
41. **4**	42. **324**	43. **48**	44. **10**	45. **1**
46. **4**	47. **2**			

Physics

48. **A**	49. **D**	50. **D**	51. **C**	52. **C**
53. **C**	54. **D**	55. **D**	56. **B**	57. **C**
58. **D**	59. **A**	60. **C**	61. **C**	62. **D**
63. **B**	64. **C**	65. **A**	66. **B**	67. **25**
68. **3**	69. **7**	70. **9**	71. **60**	72. **243**
73. **150**	74. **4**	75. **1**		

Out of syllabus

76. **A**	77. **D**	78. **C**	79. **D**	80. **1**
81. **C**	82. **C**	83. **A**	84. **A**	85. **B**
86. **A**	87. **9**	88. **3**	89. **D**	90. **9**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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